# Temporal Multi-Task Network for Audio-Based Traffic Monitoring

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## Motivation

Roadside audio offers an inexpensive alternative to camera-based traffic monitoring. Recognizing vehicles remains challenging because recordings contain background noise and overlapping vehicles. Whole clip features capture broader context, while short segments preserve local details. We introduce the **Temporal Traffic Task Network (T3N)** to combine these complementary views and jointly predict vehicle presence and vehicle type.

## Datasets

We evaluate T3N on MELAUDIS, IDMT-Traffic, MVD, and MVDA, which contain **58,111 clips** in total. The experiments use matched samples from these datasets for vehicle detection and vehicle type classification. MVDA contains augmented versions of MVD recordings. We split both datasets by source recording to prevent related clips from appearing in training and test sets.

**MELAUDIS & IDMT-Traffic:** real-world roadside recordings with urban background noise and vehicle pass-bys.

**MVD & MVDA:** MVDA is an acoustically augmented version of MVD, not an independent set of recordings; following the paper, data are split by source recording.

**Table 1.** Controlled benchmark datasets.

| Dataset | Clips/rows | Tasks |
|---|---|---|
| MELAUDIS | 19,460 | Detection, type, traffic state, count bins |
| IDMT-Traffic | 17,506 | Detection, type |
| MVD | 4,229 | Detection, type |
| MVDA | 16,916 | Detection, type |
| **Total** | **58,111** | |

## Partial Vehicle Type Supervision

In practical deployments, annotators may be able to identify whether a vehicle is present without confidently identifying its exact type. We test this setting by retaining all vehicle presence labels while using only **10%, 25%, 50%, or 100%** of vehicle type labels. The four datasets and four annotation levels produce 16 evaluation settings.

Our framework T3N shares information between detection and type prediction. For vehicle clips without a type label, its training objective encourages the type classifier to assign probability to a vehicle class, even though the exact class is unknown. This allows the model to use clips that lack detailed annotations.

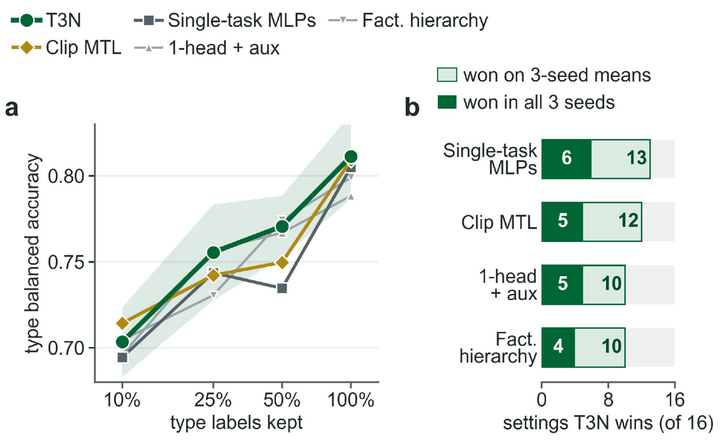



**Fig. 1.** (a) Vehicle-type balanced accuracy, mean of 4 datasets (band: T3N ± 1 SD over 3 seeds). (b) Dataset × fraction settings T3N wins, of 16.

## T3N Architecture

We propose the **Temporal Traffic Task Network (T3N)**, a lightweight multi-task framework built on frozen Audio Spectrogram Transformer (AST) and Contrastive Language-Audio Pretraining (CLAP) representations. We evaluate T3N across four traffic audio datasets using **three random seeds** and matched training and evaluation samples. Each experiment uses up to 1,500 training and 800 evaluation clips, frozen AST+CLAP features, and four temporal segments. **Balanced accuracy** is the primary metric to account for class imbalance.

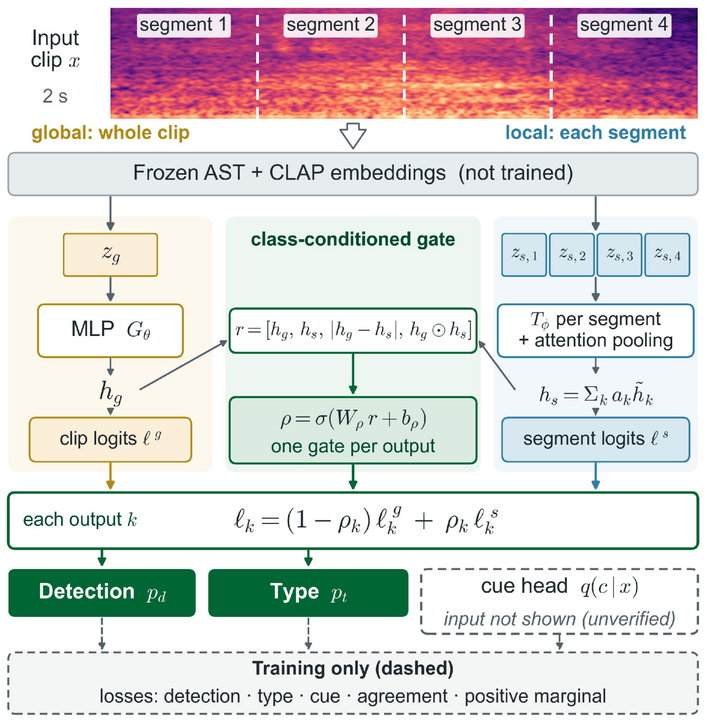



**Fig. 2.** T3N framework.

## Routing and Fusion Results

Combining global clip context with local segment improves performance. T3N outperforms **clip-only multi-task learning in 5 of 8 dataset and task settings** and **segment-only attention pooling in 7 of 8**. However, **fixed probability averaging outperforms T3N in 7 of 8 settings.** This suggests the benefit comes mainly from combining the two views.

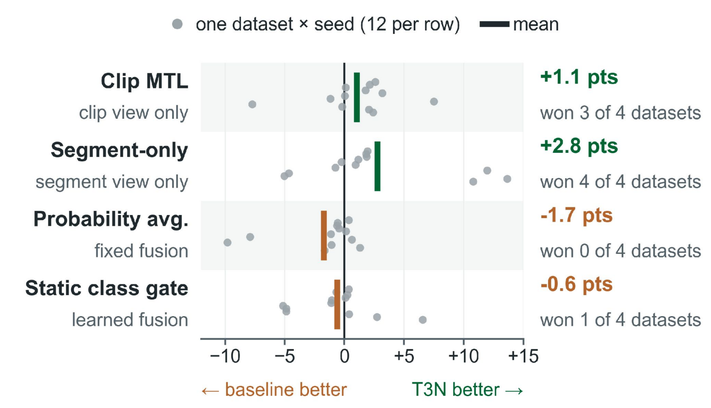



**Fig. 3.** Vehicle type balanced accuracy, T3N minus baseline; dots are the 12 paired dataset and seed differences.

## Label Hierarchy & Ablations

Adding direct detection type consistency reduced average vehicle type balanced accuracy from 0.8200 to 0.8019, a decrease of 1.81 percentage points. Average detection balanced accuracy changed only slightly, from 0.9771 to 0.9766. This result motivated omitting direct consistency from the subsequent framework.

**Table 2.** T3N ablation results for vehicle detection and vehicle type classification.

| Variant | Detection | Type | Mean |
|---|---|---|---|
| **T3N (final)** | **0.977** | **0.820** | **0.899** |
| + direct det.–type consistency | 0.977 | 0.802 | 0.889 |
| Type marginal only | 0.968 | 0.797 | 0.882 |

## Class-Conditioned Temporal Routing

T3N learns how much each output relies on the whole recording versus short segments. Gate values vary across datasets and classes, but **only some patterns are consistent across all three seeds**: for example, IDMT detection relies on the whole clip, MVDA detection on segments, and car relies on segments in MELAUDIS but on the whole clip in IDMT.

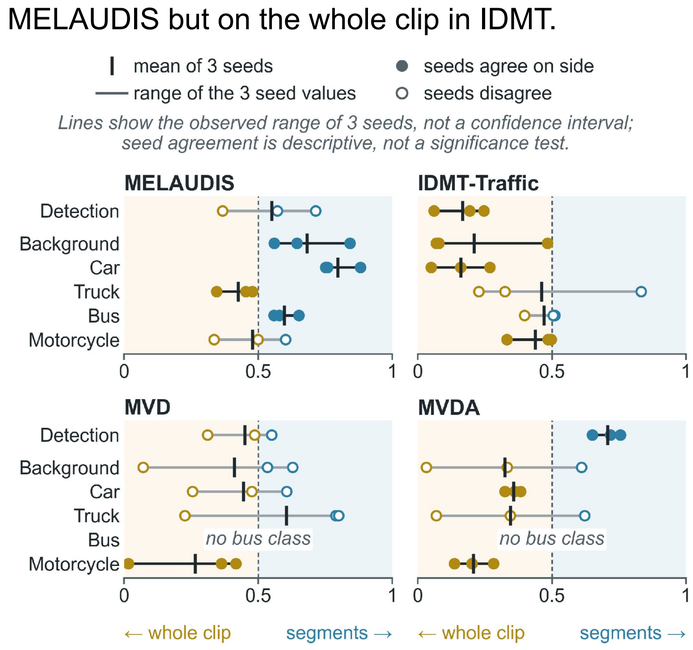



**Fig. 4.** Learned segment gate weights for each of 3 seeds (0 = whole clip, 1 = segments).

## Cross-Dataset Transfer Results

Models were trained on one dataset and evaluated on another without retraining. T3N beats logistic regression, single-task MLPs, segment-only pooling and fixed clip–segment averaging in **5 of 8 transfer settings** each, and is best for **MVD to MVDA (0.979 detection, 0.918 vehicle type)**, original vs. augmented versions of the same recordings. Between different datasets it ranks 5th-7th in three of four settings.

| | | LR | MLP | 1H | Fact. | Clip | Seg. | Avg. | T3N | T3N SD | rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **MVD** | Det. | .973 | .962 | .972 | .972 | .967 | .978 | .973 | **.979** | .001 | **1st** |
| **→ MVDA** | Type | .903 | .897 | .900 | .897 | .904 | .889 | .904 | **.918** | .007 | **1st** |
| **MVDA** | Det. | .974 | .984 | .976 | .979 | **.985** | .974 | .982 | .976 | .003 | **5th** |
| **→ MVD** | Type | .872 | .898 | .905 | .896 | .896 | **.912** | .911 | .906 | .008 | **3rd** |
| **IDMT** | Det. | .759 | .719 | .740 | .720 | .717 | **.791** | .703 | .714 | **.023** | **7th** |
| **→ MELAUDIS** | Type | .348 | .426 | .421 | .402 | .416 | .415 | .440 | **.472** | **.074** | **1st** |
| **MELAUDIS** | Det. | **.995** | .995 | .991 | .993 | .995 | .858 | .986 | .965 | **.048** | **7th** |
| **→ IDMT** | Type | .426 | .382 | .370 | .412 | .403 | **.476** | .394 | .399 | **.040** | **5th** |

colour: cell − row median (pts) ≤ −6 … 0 … ≥ +6

same recordings (MVDA = augmented MVD)
different datasets (domain shift)

Cells: target balanced accuracy (3-seed mean); bold = best in row.
T3N SD: seed SD of T3N; orange where ≥ .02.

**Fig. 5.** Cross-dataset transfer results.

## Conclusion

T3N jointly predicts vehicle presence and vehicle type by combining global clip context with local temporal information through learned class-specific gates. It **improves vehicle type balanced accuracy** over both single-view baselines on average, **without finetuning** foundation encoders, and outperforms several baselines when type labels are **limited**. Transfer results are strongest between acoustically related datasets, particularly from MVD to MVDA. Simple probability averaging remains the strongest fusion control in the routing experiment. An important next step is to compare T3N with probability averaging under partial vehicle type supervision and cross-dataset transfer.

**Source code · github.com/AcaiLab/Traffic**